

Part 1: Regulatory Audit — Which Vehicle Specs Actually Matter?

Parameter	Why It Matters Legally	Impact on Senegal (Dakar)	Impact on Morocco (Casablanca)
Year of First Reg.	Strict import age cutoff laws	Fatal Gate: Must be $\leq$ 10 years old.	Fatal Gate: Must be $\leq$ 5 years old (unless MRE retiree: $\leq$ 10 yrs).
Listing Price (CAD)	Customs valuation (CIF baseline)	Drives the 20% duty + 18% VAT base.	Drives 17.5% duty + 20% VAT base.
Fuel Type	Environmental taxes & exemptions	Marginal statistical impact.	Massive: Pure EVs are exempt from TIC; Diesels pay significantly higher TIC than Gasoline.
Engine Displacement (cc)	Puissance Fiscale (CV) bracket	Minimal (fixed tariff band).	Massive: Determines TIC excise bracket (from 25,000 MAD up to 240,000 MAD).
Vehicle Class	Customs classification (HS code)	Distinguishes light cars ($\leq$ 3.5T, 10-yr cap) from heavy trucks ($>$ 3.5T, 15-yr cap).	Distinguishes passenger cars from commercial utility vans.
Steering Layout	Road safety legislation	Fatal Gate: Must be Left-Hand Drive (LHD).	Fatal Gate: Must be Left-Hand Drive (LHD).

Note on Steering: Canadian domestic-market vehicles are 99.9% LHD. The extension should **default to LHD automatically** and only flag an alert if the title or listing explicitly mentions JDM (Japanese import) or RHD.

Part 2: How to Auto-Deduce Specs Without a VIN

When a user opens a Marketplace listing, the extension should scrape the listing title (which almost always follows the pattern: [Year] [Make] [Model], e.g., "2021 Toyota RAV4 LE"), and extract the price.

To resolve the missing technical specs (Engine CC and Fuel Type) without forcing the user to type them, integrate one of these APIs:

CarAPI.app — Best for North American Trims

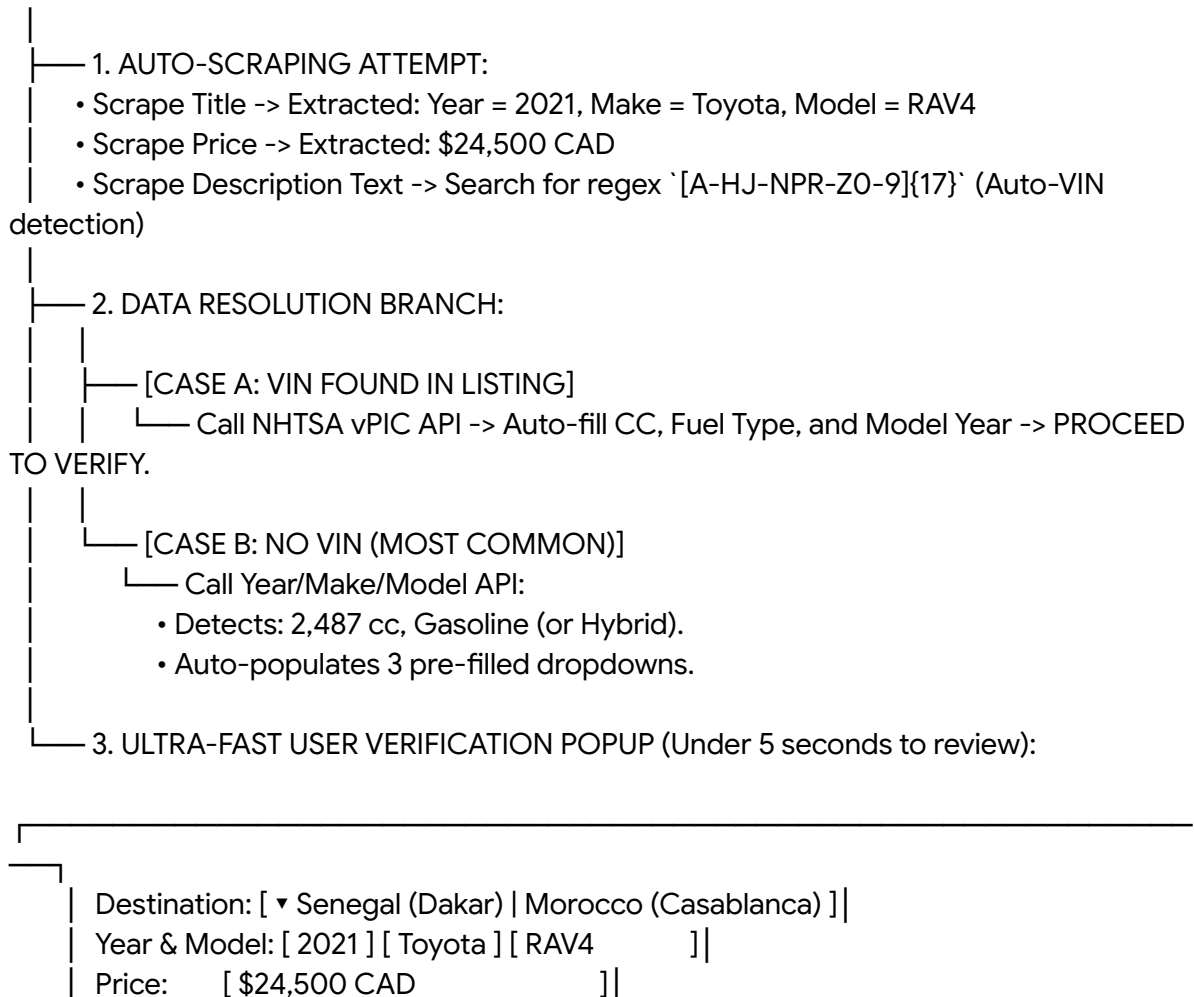
- **Endpoints:**
Year-Make-Model-Trim (YMMT) lookup. Look at the documentation (btw its only for vehicles between 2015-2020 so we will demonstrate only with these years, if we give other than between 2015-2020, say unable to get for now please input between 2015-2020).
- **What it returns:** Specific engine specs for Canadian/US market trims (e.g., distinguishing between a RAV4 2.5L Gas vs. RAV4 Hybrid).
- **Free tier:** lookup the endpoints

NHTSA vPIC API (US/Canadian Federal Database) — 100% Free Government Tool

- **Fallback VIN Decoder:** If a seller *does* put a VIN in the description, query
https://vpic.nhtsa.dot.gov/api/vehicles/DecodeVin/{VIN}?format=json. It returns all engine, displacement, and fuel variables instantly at zero cost. Make that an option also if there is a vin

Part 3: The Chrome Extension User Experience (Fast-Path Flow)

[USER CLICKS EXTENSION ON FACEBOOK MARKETPLACE]



| Engine Specs: [2,487 cc ▼] Fuel: [Gasoline ▼] |
| [CALCULATE CLEARANCE & ELIGIBILITY] |

Part 4: Complete Unified Logic Tree for Your Worker

SECTION 1: INPUT GATHERING & DEDUCTION

[Inputs Required]

- destination_country: 'SEN' or 'MOR'
- vehicle_year: YYYY (Integer)
- purchase_price_cad: Float (from listing price)
- engine_cc: Integer (Auto-deduced via API from Make/Model/Year, fallback user input)
- fuel_type: Enum ('Gasoline', 'Diesel', 'Hybrid', 'Electric')
- is_mre_retiree: Boolean (Only active if destination == 'MOR', defaults to False)

SECTION 2: REAL-TIME FX VALUATION ENGINE

- API Source: Frankfurter API (open-source) or ExchangeRate-API.
- FX Routing Logic:
 - IF destination == 'SEN':
 1. Fetch CAD to EUR mid-market spot rate.
 2. Convert to XOF via constant Euro parity:
 $CADXOF_spot = CADEUR * 655.957$
 3. Effective Bank Settlement Rate = $CADXOF_spot * (1 - 0.025)$ [2.5% retail bank spread]
 4. Customs Assessed Rate = Use static weekly official customs peg.
 - IF destination == 'MOR':
 1. Fetch CAD to MAD mid-market spot rate.
 2. Effective Bank Settlement Rate = $CADMAD_spot * (1 - 0.022)$ [2.2% retail bank spread]
 3. Customs Assessed Rate = Use static weekly Bank Al-Maghrib rate.

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SECTION 3: PASS/FAIL REGULATORY COMPLIANCE GATES

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[GATE 0: CANADIAN EXPORT ORIGIN (BOTH ROUTES)]

- CBSA CERS Check:
 - IF purchase_price_cad > 2000 CAD:
 - CERS export declaration mandatory >= 48 hours prior to vessel loading.
 - Forwarder fee: ~\$100 CAD.
- PPSA Lien Warning:
 - Reminder to user: Confirm vehicle has zero outstanding bank financing.

[GATE 1: SENEGAL CORRIDOR (PORT OF DAKAR)]

- Age Calculation: vehicle_age = Current_Year - vehicle_year
- Age Rule:
 - IF vehicle_age > 10:
 - STATUS: REJECTED (FATAL)
 - MESSAGE: "Senegalese Customs strictly prohibits passenger vehicles over 10 years old. Entry will be denied at Dakar."
 - STOP PROCESSING.
 - ELSE:
 - STATUS: ELIGIBLE.
- Mandatory Filing: BSC (Bordereau de Suivi de Cargaison) required before sailing (~\$150 CAD).

[GATE 2: MOROCCO CORRIDOR (PORT OF CASABLANCA / TANGER MED)]

- Age Calculation: vehicle_age = Current_Year - vehicle_year
- Age Rule:
 - IF is_mre_retiree == True:
 - IF vehicle_age > 10:
 - STATUS: REJECTED (FATAL)
 - MESSAGE: "MRE retirement exemption only allows vehicles up to 10 years old."
 - STOP PROCESSING.
 - ELSE:
 - STATUS: ELIGIBLE (MRE 85% abatement active).
 - ELSE (Standard Importer):
 - IF vehicle_age >= 5:
 - STATUS: REJECTED (FATAL)
 - MESSAGE: "Moroccan Customs bans standard passenger imports 5 years or older. Entry will be refused."
 - STOP PROCESSING.
 - ELSE:

STATUS: ELIGIBLE.

- Mandatory Homologation: NARSA compliance inspection required post-arrival (~\$250 CAD).

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SECTION 4: FINANCIAL & TAX VALUATION FORMULAS

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[COMMON LOGISTICS COSTS ESTIMATOR]

- Ro-Ro Ocean Freight (Montreal/Halifax to Dakar/Casablanca): ~\$2,200 CAD
- Marine Transit Insurance: 1.5% of purchase_price_cad
- CIF_CAD = purchase_price_cad + 2200 + (purchase_price_cad * 0.015)

[IF DESTINATION == SENEGAL]

1. CIF_XOF = CIF_CAD * CADXOF_rate
2. Customs Duty (DD) = 20% * CIF_XOF
3. Statistical Tax (RS) = 1% * CIF_XOF
4. Regional Levies (PCS + PCI + COSEC) = 1.7% * CIF_XOF
5. Duty Base for VAT = CIF_XOF + DD + RS + (1.7% * CIF_XOF)
6. VAT (TVA) = 18% * Duty Base
7. Total Senegal Taxes (XOF) = DD + RS + (1.7% * CIF_XOF) + VAT
8. Estimated Total Taxes (CAD) = Total Senegal Taxes / CADXOF_rate
9. Add Dakar Port Transit & Clearing Fees: ~\$1,000 CAD

[IF DESTINATION == MOROCCO]

1. CIF_MAD = CIF_CAD * CADMAD_rate
2. Customs Import Duty (DD):
 - IF is_mre_retiree == True:
DD = (17.5% * CIF_MAD) * 0.15 [85% abatement]
 - ELSE:
DD = 17.5% * CIF_MAD
3. Parafiscal Tax (TPI) = 0.25% * CIF_MAD
4. Domestic Consumption Tax (TIC) calculation based on engine_cc & fuel_type:
 - IF fuel_type == 'Electric':
TIC = 0 MAD
 - ELSE IF engine_cc < 1500:
TIC = (fuel_type == 'Diesel') ? 30,000 MAD : 25,000 MAD
 - ELSE IF engine_cc BETWEEN 1500 AND 1999:

TIC = (fuel_type == 'Diesel') ? 48,000 MAD : 40,000 MAD

ELSE IF engine_cc BETWEEN 2000 AND 2499:

TIC = (fuel_type == 'Diesel') ? 72,000 MAD : 60,000 MAD

ELSE (>= 2500 cc):

TIC = (fuel_type == 'Diesel') ? 120,000 MAD : 100,000 MAD

5. Value Added Tax (TVA) compounds on TIC:

$TVA = 20\% * (CIF_MAD + DD + TPI + TIC)$

6. Total Morocco Taxes (MAD) = DD + TPI + TIC + TVA

7. Estimated Total Taxes (CAD) = Total Morocco Taxes / CADMAD_rate

8. Add Casablanca Transit, NARSA Homologation & Port Fees: ~\$1,200 CAD